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Inventor(s)	Lin; Ming-Cheng

Reusable holding component for heatsink

Abstract

Disclosed example reusable holding components include: a frame with a fastener receiving opening surrounding by a fastener guiding wall and extending from a first surface of the frame to a second surface of the frame; at least two pins located on opposite sides of the fastener receiving opening; each of the pins having a first end disposed on the second surface of the frame, each pin extending away from the second surface to a second end of each pin, wherein the second end of each pin includes at least two elongated segments with hooks disposed on a head portion of each of the at least two elongated segments; and a heatsink having a top surface and opposite bottom surface and the top surface connected to the reusable holding component by the at least one pin.

Inventors:	**Lin; Ming-Cheng (Kaohsiung, TW)**
Applicant:	**Illinois Tool Works Inc.** (Glenview, IL)
Family ID:	**1000008819310**
Assignee:	**Illinois Tool Works Inc. (Glenview, IL)**
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Primary Examiner: Wu; Jerry

Attorney, Agent or Firm: McAndrews, Held & Malloy. Ltd.

Background/Summary

FIELD OF THE DISCLOSURE
(1) The present disclosure is related to a reusable holding component for a heatsink.
BACKGROUND
(2) In a conventional computer assembly, heat dissipation is performed by attaching a heatsink on the top of a central processing unit (CPU). The heatsink may include a plurality of cooling fins on the top or sides of the heatsink to conduct heat from the top of the CPU to the bottom of the heatsink. Heat will be dissipated by the cooling fins.
(3) The heatsink can be attached to a frame of a computer using a variety of attachment mechanisms, such as clamps, screws, and other hardware. One conventional fastener includes a screw with a head, a tapered ring, and a sealing ring. The tapered ring may be disposed on the screw below the head of the screw, and may have a tapered lower peripheral surface. The sealing ring may be disposed on the screw and engaging the tapered lower peripheral surface of the tapered ring. The conventional fastener may be disposed within a bore in the heatsink and threaded into a load frame for securing the heatsink to the load frame.
(4) However, the structure of the conventional fastener for securing the heatsink to the load frame is complex. In addition, securing the heatsink by the conventional fastener to the load frame might take multiple steps and might destroy the structure of either the fastener or the load frame. Therefore, there is a need for improving the structure of the fastener and the heatsink.
SUMMARY
(5) Embodiments of the present disclosure generally relate to a reusable holding component for a heatsink, a heat transfer device, and an electronic device with a heatsink. In one embodiment, a reusable holding component is provided. The reusable holding component may comprise a frame with a fastener receiving opening extending from a first surface of the frame to a second surface of

the frame, and at least one pin disposed on and extending away from the second surface of the frame, wherein the at least one pin includes at least two elongated segments with hooks disposed on a head portion of each of the at least two elongated segments.
(6) In another embodiment, a heat transfer device is provided. The heat transfer device may comprise a reusable holding component comprising a frame with a fastener receiving opening extending from a first surface of the frame to a second surface of the frame, and at least one pin disposed on and extending away from the second surface of the frame, wherein the at least one pin includes at least two elongated segments with hooks disposed on a head portion of each of the at least two elongated segments, and a heatsink connected to the heatsink by the at least one pin, and the heatsink has a fastener receiving opening aligned with the fastener receiving opening of the frame, and at least one guiding opening corresponding to the at least one pin of the reusable holding component.
(7) In yet another embodiment, an electronic device is provided. The electronic device may comprise an electronic component supported by a base, a heatsink connected to the base for dissipating heat from the electronic component, a reusable holding component connected the heatsink comprising a frame with an opening extending from a first surface of the frame to a second surface of the frame, and at least one pin disposed on and extending away from the second surface of the frame, wherein the at least one pin includes at least two elongated segments with hooks disposed on a head portion of each of the at least two elongated segments, wherein the heatsink has a fastener receiving opening aligned with the fastener receiving opening of the frame, and at least one guiding opening corresponding to the at least one pin of the reusable holding component, wherein the base has a fastener receiving opening, wherein the fastener receiving opening of the base is aligned with the fastener receiving opening of the frame and the fastener receiving opening of the heatsink a fastener configured to secure the reusable holding component and the heatsink to the base through the fastener receiving opening of the frame, the fastener receiving opening of the heatsink, and the fastener receiving opening of the base.
(8) The above and other embodiments of the present disclosure are described in more details in the following contexts.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) So that the manner in which the above described features of the present disclosure can be understood, a more specific description of the disclosure, briefly summarized above, may be had by reference to embodiments, some of which are illustrated in the appended drawings. However, the appended drawings illustrate only exemplary embodiments of this disclosure. It is to be understood that the disclosure may admit to other equally effective embodiments, and therefore the appended drawings should not be considered as limiting the scope of the present disclosure.
(2) FIG. **1** illustrates a schematic view of a reusable holding component according to an embodiment of the present disclosure.
(3) FIG. **2** illustrates another schematic view of the reusable holding component as shown in FIG. **1**.
(4) FIG. **3** illustrates a schematic view of a pedestal of a heatsink according to an embodiment of the present disclosure.
(5) FIG. **4** illustrates another schematic view of the pedestal of the heatsink as shown in FIG. **3**.
(6) FIG. **5** illustrates a cross-section view of the pedestal of the heatsink taking from line A-A of FIG. **3**.
(7) FIG. **6** illustrates a schematic view of a reusable holding component in combination with a pedestal of a heatsink according to an embodiment of the present disclosure.

(8) FIG. **7** illustrates another schematic view of the c reusable holding component in combination with the pedestal of the heatsink as shown in FIG. **6**.
(9) To facilitate understanding, identical reference numerals have been used, where possible, to designate identical elements that are common in the figures. For the sake of clarity, the various embodiments shown in the figures are not necessarily drawn to scale and are illustrative representations.

DETAILED DESCRIPTION

(10) Now the embodiments of the present disclosure will be described in details with reference to the drawings.
(11) FIG. **1** illustrates a schematic view of a reusable holding component **100** according to an embodiment of the present disclosure, and FIG. **2** illustrates another schematic view of the reusable holding component **100** of FIG. **1**. As shown in FIGS. **1** and **2**, the reusable holding component **100** may include a frame **102** with a fastener receiving opening **106** and at least one pin **104**. The fastener receiving opening **106** is formed in the frame **102** and extends from a top surface of the frame **102** to a bottom surface **118** of the frame **102**. The at least one pin **104** is disposed on and extending away from the bottom surface **118** of the frame **102**. The at least one pin **104** may pass through at least one corresponding hole of a heatsink, such that the reusable holding component **100** may be attached to the heatsink. As such, the reusable holding component **100** may hold the heatsink in place, so as to facilitate the subsequent mounting of the fastener into the base.
(12) The reusable holding component **100** may further include a fastener guiding wall **108**. The fastener guiding wall **108** is disposed on and extending away from the top surface **116** of the frame **102**. In order to guide the placement of the fastener, the fastener guiding wall **108** may surround the fastener receiving opening **106** of the frame **102**. In such way, the fastener guiding wall **108** will direct the fastener into the fastener receiving opening **106** of the frame **102** during installation of the fastener.
(13) As shown in FIG. **2**, the at least one pin **104** may include at least two elongated segments **112** with hooks **114**. The hooks **114** are disposed on a head portion of each of the at least two elongated segments **112**, and the hooks **114** are tapered off to a point of the head portion of each of the at least two elongated segments **112**. The at least one pin **104** may extend vertically from the bottom surface **118** of the frame **102**. The at least two elongated segments **112** are parallel with each other, and there is a gap **110** between the at least two elongated segments. In some embodiments, the at least two elongated segments **112** may be angled with each other.
(14) The frame **102** may have a rectangular shape, and at least two pins **104** may be disposed substantially at diagonal corners on the bottom surface **118** of the frame **102**. When the reusable holding component **100** is attached to the heatsink by the at least two pins **104**, the at least two pins **104** at diagonal corners may ensure that the reusable holding component **100** is attached to the heatsink securely.
(15) FIG. **3** illustrates a schematic view of a pedestal **300** of a heatsink according to an embodiment of the present disclosure, FIG. **4** illustrates another schematic view of the pedestal **300** of the heatsink of FIG. **3**, and FIG. **5** illustrates a cross-section view of the pedestal of the heatsink of FIG. **3**.
(16) Typically, heatsinks are formed with fins, pins or other similar structures to increase the surface area of the heatsink and thereby enhance heat dissipation. Heatsinks are typically formed of metals, such as copper or aluminum. The heatsinks are attached to a base using a variety of attachment mechanisms, such as clamps, screws, and other hardware for dissipating heat from an electronic component, and the electronic component is supported by the base to contact with heatsinks.
(17) As shown in FIGS. **3** and **4**, the pedestal **300** of the heatsink has a top surface **306** and a bottom surface **308**, and has a fastener receiving opening **302** and at least one guiding opening **304**. The fastener receiving opening **302** of the pedestal **300** of the heatsink and the at least one guiding

opening **304** of the pedestal **300** of the heatsink may extend from the top surface **306** of the pedestal **300** of the heatsink to the bottom surface **308** of the pedestal **300** of the heatsink. The at least one guiding opening **304** of the pedestal **300** of the heatsink corresponds to at least one pin of the reusable holding component as shown in FIGS. **1** and **2**.

(18) In some embodiment, at least two guiding openings **304** are disposed substantially at diagonal corners on the pedestal **300** of the heatsink. At least two pins of the reusable holding component at the corresponding diagonal corners may pass through the at least two guiding openings **304** to attach the reusable holding component to the pedestal **300** of the heatsink, to ensure that the reusable holding component is attached to the pedestal **300** of the heatsink securely.

(19) As shown in FIG. **5**, the at least one guiding opening **304** of the pedestal **300** of the heatsink may extend from the top surface **306** of the pedestal **300** of the heatsink to the bottom surface **308** of the pedestal **300** of the heatsink in a stepped way. The diameter of the at least one guiding opening **304** of the pedestal **300** of the heatsink at the top surface **306** of the pedestal **300** of the heatsink may be smaller than the diameter of at least one guiding opening **304** of the pedestal **300** of the heatsink at the bottom surface **308** of the pedestal **300** of the heatsink. In such way, the hooks of the at least one pin of the reusable holding component as shown in FIGS. **1** and **2** may hook within the at least one guiding opening **304** of the pedestal **300** of the heatsink to prevent the reusable holding component from removing the pedestal **300** of the heatsink.

(20) In some embodiments, the at least one guiding opening **304** of the pedestal **300** of the heatsink may extend from the top surface **306** of the pedestal **300** of the heatsink to the bottom surface **308** of the pedestal **300** of the heatsink in a straight way, such that the hooks of the at least one pin of the reusable holding component may hook onto the bottom surface **308** of the pedestal **300** of the heatsink to prevent the reusable holding component from falling out of the pedestal **300** easily.

(21) Still referring to FIG. **5**, the fastener receiving opening **302** of the pedestal **300** of the heatsink extends from the top surface **306** of the pedestal **300** of the heatsink to the bottom surface **308** of the pedestal **300** of the heatsink in a stepped way. The diameter of the fastener receiving opening **302** of the pedestal **300** of the heatsink at the top surface **306** of the pedestal **300** of the heatsink is larger than the diameter of the fastener receiving opening of the pedestal **300** of the heatsink at the bottom surface **308** of the pedestal **300** of the heatsink. In such way, the operator may choose the size of the fastener to fit the diameter of the fastener receiving opening **302** of the pedestal **300** of the heatsink at the bottom surface **308** of the pedestal **300** of the heatsink, and may operate the fastener to pass through the fastener receiving opening **302** of the pedestal **300** of the heatsink easily.

(22) FIG. **6** illustrates a schematic view of a reusable holding component **602** in combination with a pedestal **620** of a heatsink according to an embodiment of the present disclosure, and FIG. **7** illustrates another schematic view of the combination of the reusable holding component **602** and the pedestal **620** of the heatsink of FIG. **6**.

(23) Similar to the reusable holding component of FIG. **1**, the reusable holding component **602** may include a frame **604** with a fastener receiving opening **606**, at least one pin **608**, and a fastener guiding wall **610**. The fastener receiving opening **606** extends from a top surface of the frame **604** of the reusable holding component **602** to a bottom surface of the frame **604** of the reusable holding component **602**. The at least one pin **608** is disposed on and extending away from the bottom surface of the frame **604** of the reusable holding component **602**. The fastener guiding wall **610** is disposed on and extending away from the top surface of the frame **604**. The fastener guiding wall **610** may surround the fastener receiving opening **608** of the frame **604**.

(24) Similar to the pedestal of the heatsink of FIG. **3**, the pedestal **620** of the heatsink has a top surface **626** and a bottom surface **628**. Further, a fastener receiving opening **622** and at least one guiding opening **624** can be formed in the pedestal **620**. The fastener receiving opening **622** of the pedestal **620** of the heatsink and the at least one guiding opening **624** of the pedestal **620** of the heatsink may extend from the top surface **626** of the pedestal **620** of the heatsink to the bottom

surface **628** of the pedestal **620** of the heatsink.

(25) As shown in FIGS. **6** and **7**, the reusable holding component **602** is attached to the pedestal **620** of the heatsink by the at least one pin **608**. The bottom surface **628** of the frame **604** of the reusable holding component **602** may engage with the top surface **626** of the pedestal **620** of the heatsink. A fastener receiving opening **624** of the pedestal **620** of the heatsink aligns with the fastener receiving opening **606** of the frame **604** of the reusable holding component **602**, and at least one guiding opening **624** of the pedestal **620** of the heatsink corresponds to the at least one pin **608** of the reusable holding component **602**.

(26) The at least one pin **608** may include at least two elongated segments with hooks. The hooks are disposed on a head portion of each of the at least two elongated segments, and the hooks extend laterally from the head portion of each of the at least two elongated segments and are tapered off to a point of the head portion of each of the at least two elongated segments. There is a gap between the at least two elongated segments. The reusable holding component **602** may be placed on the top surface **626** of the pedestal **620** of the heatsink, and then the at least one pin **608** may pass through the at least one guiding opening **624** of the pedestal **620** of the heatsink by decreasing the gap between the at least two elongated segments. After passing through the at least one guiding opening **624** of the pedestal **620** of the heatsink, the gap between the at least two elongated segments restores and the reusable holding component **602** is roughly fixed on the pedestal **620** of the heatsink. On the other hand, the operator may press the hooks of the at least one pin **608** to decrease the gap between the at least two elongated segments, such that the at least two elongated segments as well as the hooks of the reusable holding component **602** may retreat from the at least one guiding opening **624** of the pedestal **620**. Consequently, the heatsink can be removed easily by the reusable holding component **602** from the pedestal **620** of the heatsink. The installation and the removal of the heatsink as described above will not infect the structure of the reusable holding component **602**. As such, the heatsink of the present disclosure can be installed on different pedestals, and can be replaced and used for many times.

(27) In some embodiments, the pedestal **620** of the heatsink with the reusable holding component **602** can be attached to a base (not shown) with a fastener. An electronic component (e.g. the CPU, microprocessor, or the likes) is supported by the base, and the heatsink attached to the base is used for dissipating heat from the electronic component. In some embodiments, the fastener (e.g. a screw with a head portion) may be disposed within the fastener receiving opening **606** of the frame **604** of the reusable holding component **602** and the fastener receiving opening **622** in the pedestal **620** of the heatsink and threaded into the base for securing the heatsink to the base.

(28) In some embodiments, the base may have a fastener receiving opening for receiving the fastener. The fastener may pass through the fastener receiving opening **606** of the frame **604** of the reusable holding component **602**, the fastener receiving opening **622** in the pedestal **620** of the heatsink, and the fastener receiving opening of the base. The fastener may include a machine screw with a head portion and a machine nut to secure the pedestal **620** of the heatsink with the reusable holding component **602** to the base.

(29) Although the disclosure herein has been described with reference to particular embodiments, those skilled in the art will understand that the embodiments described are merely illustrative of the principles and applications of the present disclosure. It will be apparent to those skilled in the art that various modifications and variations can be made to the method and apparatus of the present disclosure without departing from the spirit and scope of the disclosure. Thus, the present disclosure can include modifications and variations that are within the scope of the appended claims and their equivalents.

Claims

1. A reusable holding component, comprising: a frame with a fastener receiving opening surrounding by a fastener guiding wall and extending from a first surface of the frame to a second surface of the frame; at least two pins located on opposite sides of the fastener receiving opening; each of the pins having a first end disposed on the second surface of the frame, each pin extending away from the second surface to a second end of each pin, wherein the second end of each pin includes at least two elongated segments with hooks disposed on a head portion of each of the at least two elongated segments; and a heatsink having a top surface and opposite bottom surface and the top surface attached to the reusable holding component by the at least one two pins, wherein the heatsink has a fastener receiving opening aligned with the fastener receiving opening of the frame, and at least one guiding opening corresponding to each pin of the reusable holding component extends from the top surface to the bottom surface in a stepped way, the diameter of each guiding opening on the top surface is smaller than the diameter of the guiding opening on the bottom surface, and wherein the hooks are received between the top surface and the bottom surface when the bottom surface is attached to a heat generator by a fastener through the fastener receiving opening.
2. The reusable holding component according to claim 1, wherein the at least one pin is vertically extended from the second surface of the frame.
3. The reusable holding component according to claim 1, wherein the frame has a rectangular shape, and at least two pins are disposed at diagonal corners on the second surface of the frame.
4. The reusable holding component according to claim 1, wherein the at least two elongated segments are parallel with each other, and there is a gap between the at least two elongated segments.
5. A heat transfer device, comprising: a reusable holding component comprising: a frame with a fastener receiving opening surrounding by a fastener guiding wall and extending from a first surface of the frame to a second surface of the frame; and at least two pins located on opposite sides of the fastener receiving opening, each of the pins having a first end disposed on the second surface of the frame, each pin extending away from the second surface to a second end of each pin, wherein the second end of each pin includes at least two elongated segments with hooks disposed on a head portion of each of the at least two elongated segments; and a heatsink having a top surface and opposite bottom surface, the top surface attached to the reusable holding component by the at least two pins, wherein the heatsink has a fastener receiving opening aligned with the fastener receiving opening of the frame, and one guiding opening corresponding to each pin of the reusable holding component extends from the top surface to the bottom surface in a stepped way, wherein the diameter of each guiding opening on the top surface is smaller than the diameter of the guiding opening on the bottom surface, and the hooks are received between the top surface and the bottom surface when the bottom surface is attached to a heat generator by a fastener through the fastener receiving opening.
6. The heat transfer device according to claim 5, wherein the at least one pin is vertically extended from the second surface of the frame.
7. The heat transfer device according to claim 5, wherein the frame has a rectangular shape, and at least two pins are disposed at diagonal corners on the second surface of the frame.
8. The heat transfer device according to claim 5, wherein the fastener receiving opening of the heatsink extends from a first surface of the heatsink to a second surface of the heatsink in a stepped way, and a first diameter of the fastener receiving opening of the heatsink at the first surface of the heatsink is smaller than a second diameter of the fastener receiving opening of the heatsink at the second surface of the heatsink.
9. The heat transfer device according to claim 5, wherein the at least two elongated segments are parallel with each other, and there is a gap between the at least two elongated segments.
10. The heat transfer device according to claim 5, wherein the fastener receiving opening of the

heatsink extends from a first surface of the heatsink to a second surface of the heatsink in a stepped way.

11. The heat transfer device according to claim 5, wherein the fastener receiving opening of the heatsink extends from a first surface of the heatsink to a second surface of the heatsink in a straight way.